From eDNA composition to ecosystem function: predicting recurrent protist communities and primary production in a changing Arctic

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Abstract

Predicting microbial community dynamics and functioning is essential to understanding ecosystem responses to global change. This is critical in the Arctic, where warming and sea-ice retreat reshape marine communities, yet protist diversity revealed by eDNA complicates identifying the drivers of community composition. Here, we analyse ~900 eDNA samples collected over 15 years in Fram Strait, spanning ice-covered Polar to ice-free Atlantic waters and >1,000 taxa. We present a diffusion-map-based approach for community distribution modelling that represents communities in a low-dimensional space, identifies recurrent community types, and applies machine learning to estimate their occurrence probabilities from environmental conditions. The approach resolved eight recurrent types and coherent compositional changes along low-dimensional trajectories, from low-productivity under-ice communities to surface bloom and deep heterotrophic communities. Surface succession progressed from early-summer autotroph-rich bloom states to late-season assemblages enriched in heterotrophs, parasites, and recyclers, while deeper communities were further enriched in parasites and recyclers. Temperature, day of year, and depth were the strongest predictors of community type. Despite the compositional nature of eDNA data, relative abundances of the identified indicator taxa predicted absolute chlorophyll-a concentrations. Thus, our approach identifies recurrent ecological states, their environmental predictors, and links between taxonomic composition and ecosystem productivity.

Introduction

Eukaryotic marine microbes are essential to the functioning of marine ecosystems and contribute substantially to energy flow and nutrient cycling. Their composition and productivity are a function of environmental conditions and biological interactions[1]. On a global scale, numerous lines of evidence support the idea that environmental gradients explain substantial spatial and temporal variation in marine communities, while the relative importance of individual drivers varies among taxonomic groups, habitats and regions[2–4]. The observation of recurring patterns of microbial community composition in marine time series[5–7] suggests that marine microbial community composition can be predicted from environmental conditions[8], providing a basis for using microbes as indicators of expected shifts in marine ecosystem structure and functioning in response to global change. Developing such scenarios is particularly important in the context of marine management and conservation in the Arctic Ocean, which is rapidly warming, with far-reaching consequences for marine ecosystems[9,10].

The long-term ecological research site (LTER) HAUSGARTEN, established in Fram Strait in 1999, provides a natural laboratory for observing responses of Arctic marine ecosystems to climate-driven changes in temperature, hydrography and sea-ice conditions, from the surface ocean to the deep sea and across levels of biological organisation from microbes to mammals[11,12]. As the major gateway between the North Atlantic and the central Arctic Ocean, Fram Strait covers a pronounced environmental gradient. Warm, saline Atlantic waters flow northwards in the eastern Fram Strait via the West Spitsbergen Current (WSC), whereas cold polar waters and sea ice are transported southwards in the west via the East Greenland Current (EGC), creating a transition from relatively warm, ice-free Atlantic to cold, ice-covered polar conditions[13]. LTER HAUSGARTEN spans this gradient, with stations distributed across the WSC, EGC, and intervening marginal ice zone (MIZ) (Fig. 1), providing a natural system for investigating how microbial communities respond to changing hydrographic and sea-ice conditions[11,14].

Protist communities and primary production in Fram Strait are shaped by interacting gradients in sea ice, light, stratification, temperature and nutrient supply. Seasonal stratification, driven by sea-ice melt in the MIZ and surface warming in the West Spitsbergen Current (WSC), initiates early-summer blooms[15,16]. Winter mixing resets the system, after which microbial succession and the timing of the spring-bloom diverge significantly across the east–west ice and hydrographic gradient[15,17]. Atlantic waters support strongly seasonal communities dominated by primary producers such as *Phaeocystis pouchetii, Chaetoceros socialis*, or *Pseudonitzschia spp.* whereas Arctic waters support cold-adapted taxa, including ice-associated diatoms such as *Fragilariopsis spp.* with weaker seasonality[17,18]. These horizontal gradients are coupled to pronounced vertical succession: photoautotrophs dominate the summer photic zone, while heterotrophic and mixotrophic taxa become increasingly important with depth, accompanied by rising microbial diversity[18,19]. Towards autumn and winter, the fraction of

heterotrophs, parasites and recyclers increases throughout the water column, marking a seasonal shift from production to microbial recycling[17].

Despite substantial progress in characterising Fram Strait hydrography hydrography[16,20] and microbial communities[12], a predictive understanding of how their interactions drive ecosystem change remains elusive. Three methodological challenges limit conventional linear approaches. First, environmental drivers are strongly collinear, for example, sea-ice melt covaries with light and temperature[21], while depth covaries with nutrient and light availability[22,23]. These collinearities complicate the attribution of community variation to individual environmental drivers. Second, ecological responses are often nonlinear[24], with peak chlorophyll concentrations and distinct community states emerging under intermediate environmental conditions that linear models may overlook[25]. Third, fixed geographic classifications, such as EGC versus WSC, can obscure the dynamic nature of ecological boundaries. Apparent community shifts within predefined regions may instead reflect the spatial displacement of ecological states as sea ice retreats, rather than local community reorganization. Together, these limitations call for approaches that resolve nonlinear responses, disentangle interacting environmental drivers and capture the spatial displacement of ecological states. Given the high diversity of microbial eukaryotes and their complex responses to multiple shifting gradients, it remains extremely difficult to interpret ecosystem-level changes in a highly variable and changing environment. This raises a critical question: despite their high taxonomic diversity, can the temporal and spatial dynamics of Arctic protist communities be distilled into low-dimensional ecological trajectories?

To address these challenges, we analyse a 15-year eDNA time series of eukaryotic microbial communities collected from June to September between 2009 and 2023 across 21 repeatedly sampled LTER HAUSGARTEN stations[12]. The dataset comprises nearly 900 samples collected at 3–5 depths within the upper 100 m and more than 1,000 ASVs overall. This unprecedented spatial and temporal coverage captures microbial communities at a scale and complexity that challenges conventional approaches.

Community-level modelling offers an alternative to taxon-by-taxon prediction. Our approach is conceptually related to the established ‘assemble first, predict later’ strategy[26]. In our implementation, however, we used diffusion maps, a nonlinear manifold-learning method, to represent eukaryotic microbial communities in a low-dimensional space and identify recurrent community types. We then combined nonlinear machine learning with sequential feature selection to predict their occurrence probabilities from a minimal set of environmental variables and used partial dependence to characterise their environmental niches. Together, these steps provide a diffusion-map-based framework for community distribution modelling (CDM) that treats recurrent community types as coherent ecological units and links their occurrence to environmental conditions, thereby providing a predictive framework for resolving ecosystem dynamics across Fram Strait.

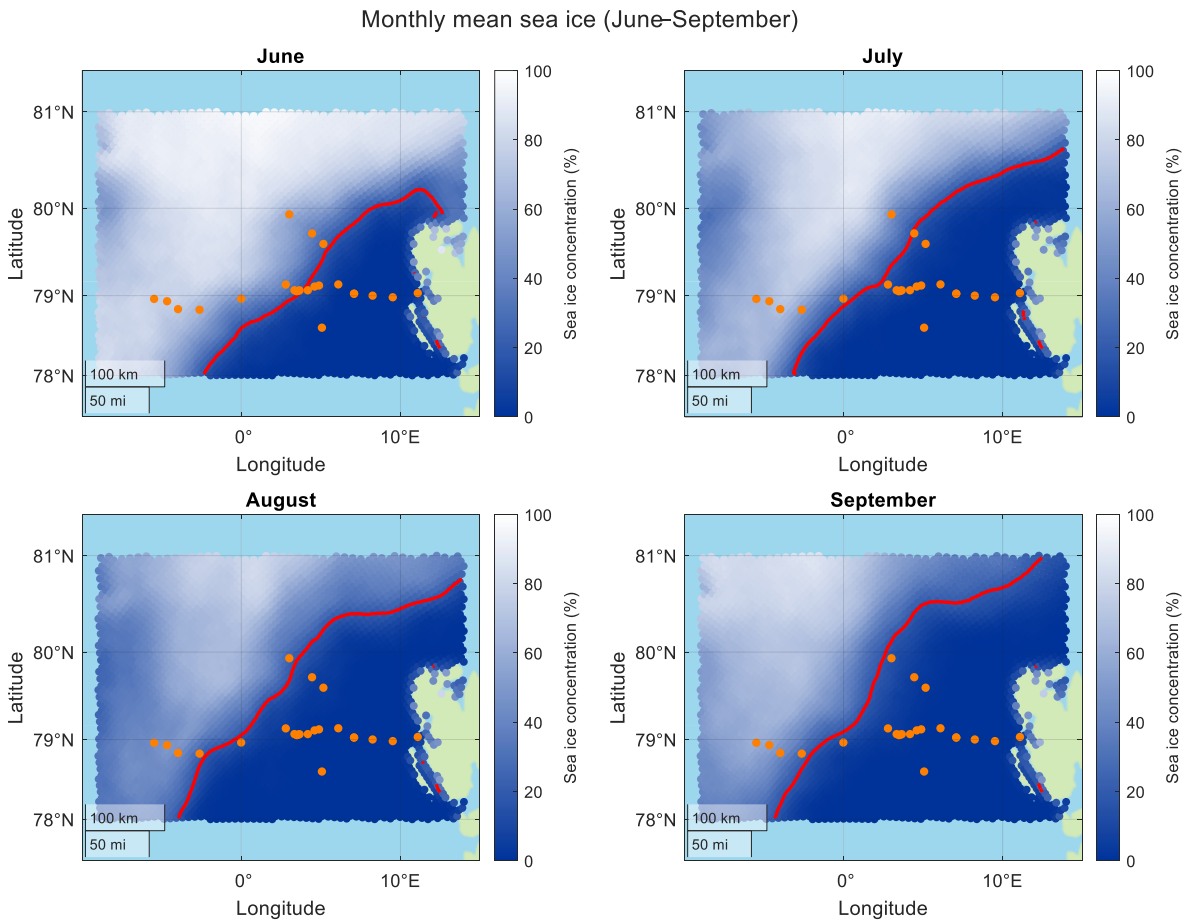


Fig. 1: Geographical location of HAUSGARTEN stations and mean sea-ice coverage (2009–2023).

Sampling stations are indicated by red circles. Background colour represents the 15-year mean sea-ice concentration for each month. The red line marks the 20% sea ice concentration isoline, representing the mean position of the sea ice edge for the respective month.

Diffusion mapping is a nonlinear manifold-learning approach that embeds high-dimensional data in a low-dimensional space while preserving local structure[27]. We previously used this framework to infer species traits and quantify functional diversity[28,29]. Here, we shift the unit of analysis from individual species to entire communities, allowing us to resolve recurrent compositional states and their spatiotemporal dynamics. Unlike conventional ordination, diffusion maps are constructed from a network of trusted nearest-neighbour links, while weak links between dissimilar communities are discarded. This local geometry is ecologically intuitive: similarity is well constrained among neighbouring communities that share taxa, whereas distances between highly dissimilar communities are less reliable. Thus, a diffusion map connects highly dissimilar communities through sequences of intermediate communities with gradually changing composition, allowing community

turnover to be represented along continuous ecological trajectories. We characterize the community diffusion map by its fractal dimension, identify clusters representing recurrent community types, and quantify their spatiotemporal dynamics, taxonomic composition, homogeneity and association with chlorophyll-a.

Our analysis reveals five key insights into Arctic protist community dynamics. First, summer communities across LTER HAUSGARTEN resolve into a limited set of recurrent states, with transitions following coherent, low-dimensional trajectories despite high taxonomic diversity. Second, seasonality, temperature and depth emerge as the dominant predictors of community type, whereas sea-ice cover contributes comparatively little. Third, seasonal dynamics reflect not only local succession but also potential spatial displacement of community states in response to warming and sea-ice retreat[30]. Fourth, community composition provides a high-resolution indicator of ecosystem productivity: despite the compositional nature of eDNA data, indicator taxa predict absolute chlorophyll-a concentrations and reveal contrasting contributions of primary producers, consumers, parasites and recyclers to productivity variation. Finally, one of the major productivity maxima is associated with ice-influenced eukaryotic microbial communities at temperatures around 0 °C and would be jeopardized by rising temperatures in polar waters.

Results and Discussion

Main ecological trajectories in diffusion map space

This study comprises nearly 900 eDNA samples capturing marine protist microbial biodiversity and community composition. Although eDNA provides high taxonomic resolution, the resulting >1000 ASVs pose a fundamental interpretive challenge: community dynamics cannot readily be interpreted in terms of individual taxa, functions or environmental gradients. To resolve the dominant modes of variation, we treated community composition as an integrated ecological unit and constructed a diffusion map from Jaccard distances between ASV profiles, without incorporating sampling date, depth, geographic position or any environmental variable into the embedding. Although constructed independently of predefined spatial, temporal or environmental information, the resulting community map revealed striking organisation along seasonal, vertical and longitudinal gradients (Fig. 2 a–c), emerging from community similarities alone. A sensitivity analysis further showed that these patterns were not driven by temporal or spatial autocorrelation among samples collected within the same year (Supplementary Information, year-masked diffusion map).

To quantify the complexity of community dynamics, we moved beyond analysis of eigenvalue spectra. Instead of defining dimensionality only by the number of retained diffusion-map axes, we estimated the intrinsic dimensionality of the embedding using the correlation dimension, a fractal-geometric measure of how samples occupy an embedding space[31]. We calculated this measure across the first ten diffusion-map components, making the estimate less dependent on the choice of retained axes. The

resulting correlation dimension was 2.00 ± 0.01 (Supplementary Fig. SfgDMCorrDim). Thus, despite a potential state space with >1,000 ASV-defined dimensions, observed communities occupied a low-dimensional manifold, indicating that only a small subset of possible community configurations was realised. Community change was therefore highly coordinated, following coherent trajectories rather than independent taxonomic shifts. This low dimensionality suggests that a small number of coupled ecological gradients may capture much of the large-scale organisation of plankton communities across Fram Strait.

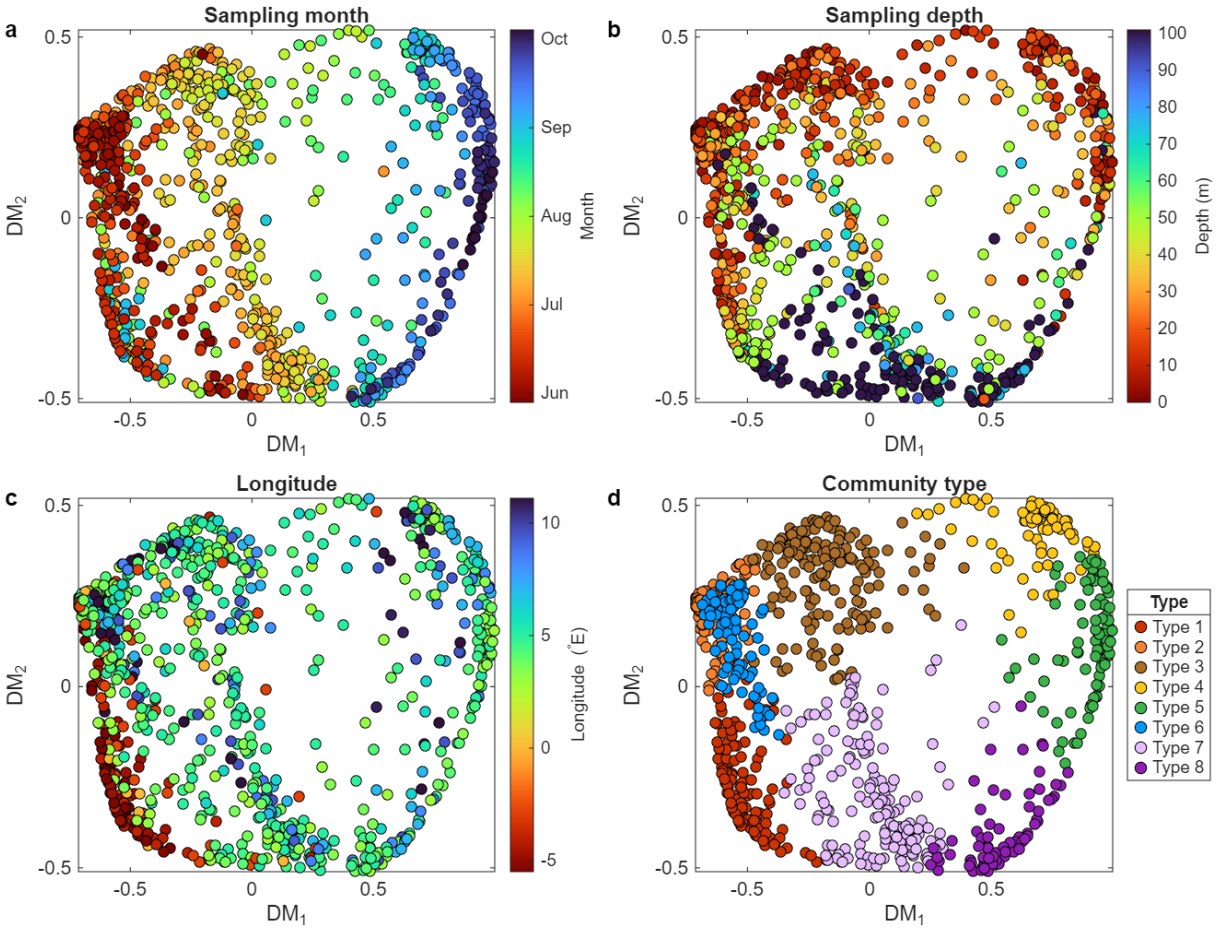


Fig. 2: Diffusion map of protist communities.

Samples are shown in the space of the first two diffusion-map coordinates (DM1 and DM2), based on pairwise Jaccard distances among samples. Points are coloured by **(a)** decimal month of sampling (rescaled from DOY), **(b)** sampling depth, **(c)** longitude and **(d)** eight community types identified using a Gaussian mixture model. The diffusion map reveals a coherent organization of community composition along seasonal (a), vertical (b) and longitudinal (c) gradients.

Correlations between diffusion-map coordinates (DMs) and environmental variables revealed three dominant ecological gradients in community structure (Supplementary Fig. SfgCorrelations). DM1 captured the main seasonal–hydrographic gradient and was strongly associated with day of year (DOY), surface photosynthetically active radiation (PAR), water temperature, sea-ice cover, and salinity, with a weaker association with mixed-layer depth (Supplementary Fig. SfgCorrelations a). This gradient captured the seasonal decline in surface PAR with DOY ($r = -0.84$), accompanied by an increase in water temperature ($r = 0.44$). Together, these variables formed a correlated seasonal–light block linked to variation in temperature, sea ice, and salinity. Thus, DM1 tracked summer progression from high-light, strongly ice-influenced conditions toward warmer late-season conditions, less impacted by sea-ice.

DM2 captured the main vertical light–biogeochemical gradient. PAR at sampling depth, chlorophyll-*a*, and major nutrients showed correlations of similar magnitude with this axis (Supplementary Fig. SfgCorrelations a), reflecting their strong covariance with depth (Supplementary Fig. SfgCorrelations b). This gradient is driven by a systematic decline in light and chlorophyll-a with depth, accompanied by increasing phosphate, silicate and dissolved inorganic nitrogen (DIN), separating shallow, well-lit, productive communities from deeper, nutrient-rich communities.

DM3 primarily captured the remaining west–east spatial hydrographic gradient, with only a weak seasonal component. Increasing DM3 values tracked the transition from colder, fresher, ice-influenced EGC waters to warmer, more saline Atlantic-influenced WSC waters. This gradient corresponded to a correlated hydrographic block comprising water temperature, satellite-derived sea-surface temperature (SST), sea-ice cover, salinity, longitude, and, more weakly, mixed-layer depth (Supplementary Fig. SfgCorrelations b).

Higher-order diffusion-map axes contained substantially less additional ecological information. DM4 was mainly associated with local DIN, whereas DM5 was primarily associated with PAR at sampling depth and DOY. DM6 and subsequent axes showed no clear additional ecological interpretation in terms of the environmental variables considered.

Thus, without using spatial, temporal, or environmental variables to construct the diffusion map, diffusion mapping independently recovered the major seasonal, vertical, and hydrographic organization of eukaryotic communities in Fram Strait[32,33,7], including the established contrast between Arctic-influenced western and Atlantic-influenced eastern waters. Even the first three DM coordinates, do not represent three independent ecological degrees of freedom: communities lie on an effectively two-dimensional surface in diffusion-map space. This organization reflects the strong covariance of marine environmental variables, as physical forcing links temperature, salinity, sea-ice dynamics, light availability, and nutrient supply across space and time[15,34]. Such covariance generates collinearity among environmental predictors and complicates attribution of ecological responses to individual variables[35]. The diffusion-map axes are therefore better interpreted as biological expressions of coupled environmental states

than as responses to individual drivers. DM1 integrates seasonal progression with hydrographic change, consistent with the joint influence of seasonality, water-mass dynamics, and sea-ice conditions on Fram Strait plankton communities[36]. DM2 captures the characteristic vertical transition from high-light, chlorophyll-rich surface waters to low-light, nutrient-rich deeper waters[37,38], whereas DM3 reflects the established hydrographic contrast between the EGC and WSC[18,39,40]

Recurrent community types

Gaussian mixture modelling in the diffusion-map space identified eight recurrent community types (Types 1–8), representing major ecological states of eukaryotic communities across seasons, depths and water masses within the upper 100 m of the HAUSGARTEN region (Table 1). Each type had a characteristic taxonomic composition and was associated with distinct seasonal, vertical and hydrographic conditions. Following the CDM approach, we used nonlinear machine learning to estimate the occurrence probability of each type from a minimal set of spatial, seasonal and environmental predictors.

Selection of environmental predictors of community type

To identify a parsimonious set of environmental variables for predicting recurrent community types in Fram Strait, we combined nonlinear multiclass support vector machine (SVM) classification with sequential feature selection. We found that community types could be predicted using only three main predictors: DOY, depth and longitude, with a cross-validated error of 0.26 ± 0.06 (mean$\pm$s.d. across folds; Supplementary Fig. SConfMat a). Replacing longitude with water temperature gave similar performance, with a cross-validated error of 0.27 ± 0.05 (Fig. SConfMat c). All four variables consistently ranked among the most informative predictors (Fig. SConfMat b, d). DOY was more informative than water temperature, consistent with DOY capturing seasonal variation in both temperature and light. This supports previous findings identifying photoperiod as a major predictor of phytoplankton community turnover[41]. In an extended model including water temperature, DOY, depth, surface PAR, nitrate and sea-ice cover, prediction improved only slightly (mean cross-validated error 0.24 ± 0.05; Supplementary Fig. SfgNicheNoDOY). Surface PAR did not replace DOY and had much lower predictor importance, suggesting that the seasonal signal was not reducible to surface PAR alone. Nitrate contributed to the slight improvement of the extended model, but its effect likely reflected nitrate uptake by primary producers rather than an independent driver of community composition (Supplementary Fig. SfgNicheNoDOY e). This interpretation is consistent with the identification of temperature and consumed nitrate as important correlates of microbial eukaryote composition in Fram Strait[42].

Predicted Probability of community types

To characterise the environmental niches of recurrent community types, we examined how their calibrated occurrence probabilities predicted by the selected models varied with DOY, depth and either longitude or water temperature. Predicted community distributions revealed pronounced seasonal and vertical zonation across Fram Strait (Fig.

3a–c). In the eastern study region, surface communities progressed from Type 2 to Type 3 in June–July, followed by Types 4 and 5 towards late summer and autumn. Type 2 emerged in early summer near 0° longitude and extended into both the WSC and ice-influenced EGC, whereas Type 6 occurred predominantly in early June and spanned the full sampled depth range. Types 7 and 8 were largely restricted to deeper WSC waters from mid-June onwards. In contrast, Type 1 dominated the ice-influenced EGC throughout most of the season and across the water column. The broad longitudinal range of the boundaries between Types 1 and 2 likely reflects interannual variability in the position of the MIZ (Ludwig and Krumpen, 2025), together with seasonal changes in water temperature. Replacing longitude with temperature produced a much narrower and more coherent boundary in temperature–month space (Fig. 3d–f), indicating that temperature provided a more informative descriptor of the environmental conditions associated with Type 1 than geographic position. This pattern is consistent with an emerging view of dynamic ecological reorganization rather than fixed geographic boundaries[43–45] . At depth, Type 1 persisted across sub-zero temperatures. At the surface, however, in early summer Type 1 was restricted to temperatures close to −2 °C, whereas Type 2 occurred at −1.5 to 1 °C, consistent with sea-ice melt conditions in the MIZ. From mid-August onward, Type 1 expanded across the sub-zero temperature range also in surface waters, replacing Type 2. Temperature–month space further revealed that most transitions between successive community types at positive temperatures were tilted rather than horizontal, indicating an interaction between thermal conditions and seasonal stage. Under colder conditions, succession was delayed or prolonged: Types 3, 4, 5 and 8 appeared later, whereas Type 7 persisted for longer. Thus, community succession was not governed by season alone[7], but emerged from the interaction between seasonal progression and the prevailing thermal–hydrographic state.

These predicted patterns are consistent with previous observations of seasonal, vertical and hydrographic structuring of microbial communities in Fram Strait[12]. Our results extend these observations by showing that the occurrence probabilities of recurrent whole-community states can be quantitatively predicted from a remarkably small set of environmental variables representing seasonality, depth and hydrography. Day of year and depth captured seasonal and vertical habitat structure, whereas temperature acted as an integrative descriptor of water-mass properties and covarying conditions associated with sea-ice melt, stratification, light availability and advection, consistent with previous Fram Strait studies linking microbial community structure to temperature and hydrography[42,46]. Sea-ice concentration and mixed-layer depth added little predictive power once these variables were accounted for, suggesting that much of their predictive information is embedded within the coupled seasonal and hydrographic gradients. This supports the broader view that complex microbial communities can be organised along a small number of dominant ecological gradients[18,47].

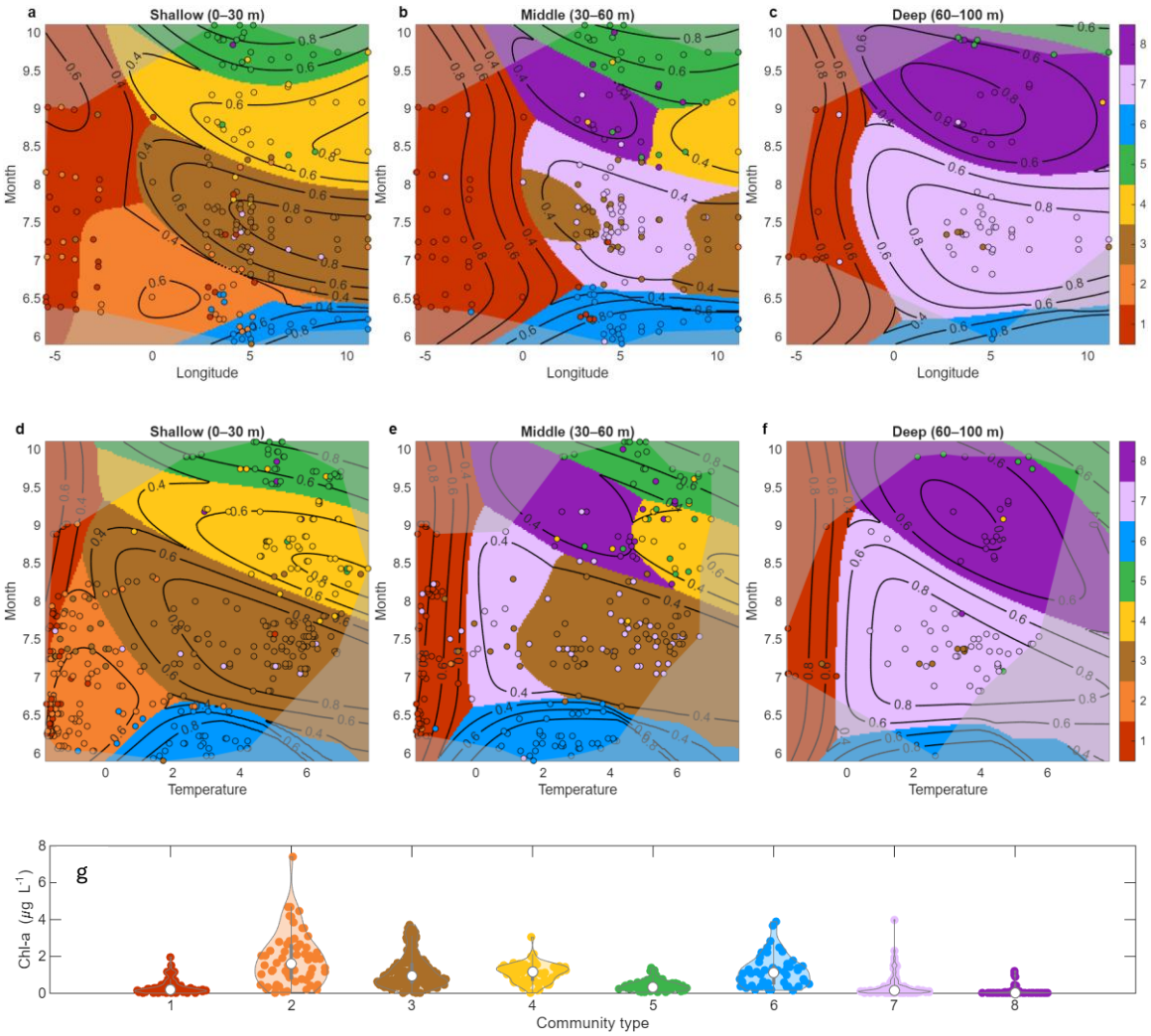


Fig. 3: Spatial-seasonal distribution and productivity of community types in Fram Strait.

a–c, Distribution of the most probable protist community type in longitude-month space averaged across shallow (0–30 m), middle (30–60 m) and deep (60–100 m) layers. **d–f,** the same but in temperature-month coordinates. Month is rescaled from DOY. Colors indicate the most probable community type, contour lines show the corresponding probability estimated with a multiclass SVM classifier, dots show the type of the original observations, the shaded outer area shows the model extrapolation range where no data is available. **g,** Chlorophyll-a concentrations across the eight community types. Points represent individual samples, violin plots show the distribution within each type, and white circles indicate the median.

Marine microbial succession is increasingly recognised as a recurrent and predictable process shaped by environmental forcing and biotic interactions[6,48,49]. Our results extend this view from temporal succession to recurrent whole-community states distributed across interacting seasonal, vertical and hydrographic gradients in a rapidly changing Arctic system. Describing these states by their environmental niches rather than as fixed geographic entities is consistent with niche-based views in which environmental filtering constrains microbial community assembly[50,51]. By combining a robust, low-dimensional representation of community composition with environmental modelling of state

occurrence, our diffusion-map-based CDM approach provides a quantitative basis for assessing how community states may be redistributed as hydrographic conditions change in Fram Strait.

The CDM framework therefore treats the whole community as the predictive unit and provides a basis for projecting how community states may redistribute in space and time under changing conditions in Fram Strait.

Productivity of community types

The community types differed markedly in chlorophyll-a standing stocks, indicating contrasting productivity states (Fig. 3 g). Type 2 had the highest chlorophyll-a concentrations, consistent with phytoplankton bloom conditions during sea-ice melt in the MIZ[15]. These conditions are consistent with enhanced phytoplankton growth promoted by increasing light availability and stratification during sea-ice retreat[48,52]. In the WSC surface layer, chlorophyll-a declined progressively from Type 2 through Types 3 and 4 to Type 5, paralleling their seasonal succession from early summer to autumn and the seasonal decline in surface irradiance (Supplementary Fig. SLightOnDoy a). Type 6 represented a second high-chlorophyll state, with concentrations comparable to those of Type 2, but occurred predominantly early in the season in warm Atlantic-influenced waters. Types 7 and 8 occupied deeper WSC layers from mid-June to September and showed the lowest chlorophyll-a concentrations. Thus, similarly high chlorophyll-a standing stocks were associated with two contrasting hydrographic settings in June: the ice-influenced MIZ and warm Atlantic waters.

Combining predicted community distributions with chlorophyll-a concentrations showed that shifts among community types were associated not only with seasonal succession and spatial displacement of community boundaries, but also with systematic changes in productivity. This was particularly evident in the western study region, where the low-chlorophyll Type 1 dominated ice-influenced EGC waters but was replaced in the surface layer during early summer by the high-chlorophyll Type 2 as sea-ice retreated. However, similarly high chlorophyll-a concentrations were also associated with Type 6 in warm Atlantic-influenced waters. Thus, transitions among community types were accompanied by systematic changes in chlorophyll-a, while similarly productive states could arise under contrasting hydrographic regimes and correspond to distinct community states, as exemplified by the MIZ-associated Type 2 and Atlantic-influenced Type 6 communities. Community types therefore provide a link between community turnover, hydrographic change and ecosystem function, offering a framework for anticipating how Arctic productivity may be reorganised as environmental conditions change.

Functionality and taxonomic composition of community types

To interpret community composition in functional terms, we developed a conceptual framework based on the ecological roles of the dominant taxonomic groups in our eDNA dataset. This framework depicts the HAUSGARTEN microbial food web as a dynamic network of phototrophs, mixotrophs, heterotrophs, recyclers and parasites that couples primary production to grazing and carbon transfer through the water column (Fig. 4). Primary production is driven largely by autotrophic and mixotrophic members of Chlorophyta, Cryptophyta, Haptophyta, Ochrophyta and Dinophyceae, which convert light, CO_2 and inorganic nutrients into organic matter that fuels microbial and higher trophic levels and contributes to vertical carbon export[53–55].

Organic matter is transferred through the food web as bacteria and small phototrophs are consumed by heterotrophic protists, including Telonemia, Picozoa, Opalozoa, Cercozoa, Choanoflagellida and Ciliophora[55,56]. These protists, in turn, can serve as prey for mesozooplankton such as copepods. Some members of Chlorophyta, Haptophyta and Dinophyceae occupy intermediate trophic positions by combining phototrophy with heterotrophic feeding, linking their ecological roles to light, nutrient availability and prey supply[57,58]. At the same time, exudation, grazing, excretion and cell lysis transfer biomass into dissolved and particulate organic-matter pools (DOM and POM), which are subsequently processed by bacteria and heterotrophic eukaryotes. In particular, heterotrophic Radiolaria contribute to the processing and vertical transport of particulate organic matter[55], while fungus-like Sagenista act as saprotrophic decomposers of algal detritus and marine snow[59]. Because these two phyla tended to occur under similar environmental conditions and participate in the processing and redistribution of organic matter, we collectively refer to them here as recyclers. Parasitic Syndiniales and Oomycetes further redirect carbon flow by infecting dinoflagellates, diatoms and other protists[60,61], channelling host-derived organic matter into DOM and POM[55]. Together, these interactions form a dynamic network of production, grazing, recycling and parasitism that provides a functional framework for interpreting seasonal changes in community composition and carbon processing.

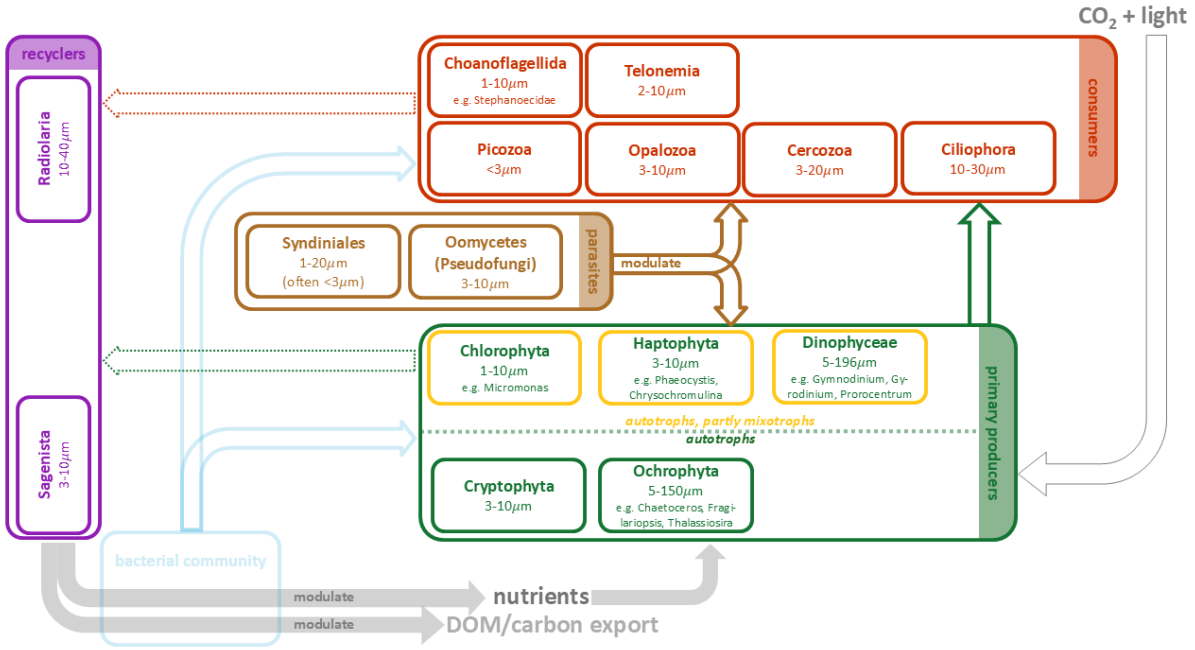


Fig. 4: Functional framework of the protist food web in Fram Strait.

The diagram illustrates the functional organization of dominant protist groups, categorized by their primary trophic roles: autotrophs (green boxes), partly mixotrophic groups (yellow boxes), parasites (brown), consumers (red), and recyclers (purple). Typical cell-size ranges are indicated for each taxonomic group. Arrows indicate trophic interactions and pathways of carbon transfer and recycling among functional groups, including grazing, lysis, and remineralization.

Taxonomic Composition of Community Types

We examined the taxonomic composition of the eight community types, focusing on phylum-level patterns and their ecological interpretation with the ultimate goal to identify indicator taxa for the community types and productivity; detailed descriptions and their inferred niches are provided in the Supplementary Information (“Description of the Eight Community Types”).

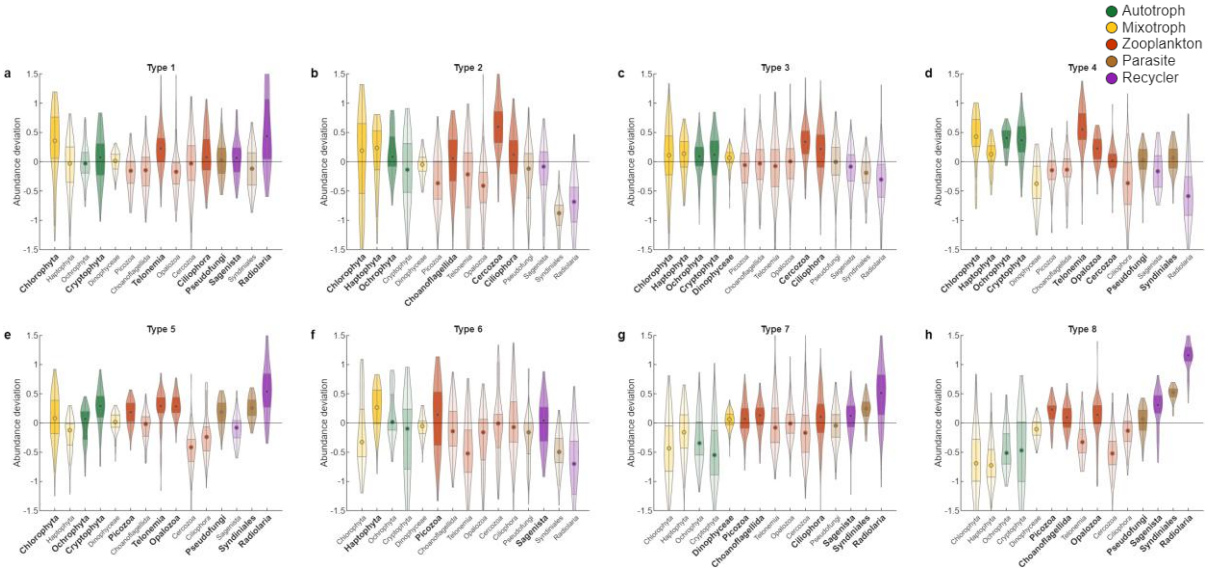


Fig. 5: Taxonomic group abundance deviation index across community types.

Distributions of the abundance deviation index D_{ij} for major taxonomic groups across community types (see Methods for definition). Values above and below zero indicate enrichment and depletion relative to the across-sample median. Taxonomic group names are shown in bold when their median deviation index is positive. Colours indicate assigned functional roles (Fig. 4).

The principal WSC surface succession (Types 2–5) progressed from bloom-associated communities enriched in primary producers, including Chlorophyta, Haptophyta and Ochrophyta, towards increasingly heterotrophic and recycling-associated assemblages (Fig. 5b–e). Early stages were enriched in Cercozoa and Ciliophora, consistent with active grazing, whereas later communities showed increasing contributions of Cryptophyta, Telonemia and Opalozoa, followed by Picozoa, Radiolaria and parasitic Syndiniales and Pseudofungi. This progression indicates a seasonal shift from autotrophic production towards heterotrophy, parasitism and organic-matter processing. Primary producers remained abundant throughout Types 2–5, but their diversity increased toward autumn, suggesting greater niche diversification as the bloom developed and declined (Fig. SFgDiv). Some taxonomic shifts were also coherent across depths; for example, Dinophyceae increased simultaneously in surface Type 3 and deep Type 7 in July, indicating shared components of community succession across the vertical gradient.

Type 6 represented a taxonomically distinct high-productivity state in Atlantic-influenced waters. Despite chlorophyll-*a* concentrations comparable to those of the MIZ-associated Type 2, its composition differed markedly, with high contributions of Haptophyta and Ochrophyta alongside Picozoa, demonstrating that similar productivity can arise from contrasting community configurations. The ice-associated Type 1 was characterized by Chlorophyta and Cryptophyta, resembling the post-bloom Type 5 assemblage, together with Telonemia, Sagenista and Radiolaria and parasitic Pseudofungi. Deep Types 7 and 8 were likewise enriched in recycling- and parasite-associated groups, including Radiolaria, Sagenista, Syndiniales and Pseudofungi,

alongside small heterotrophic consumers such as Picozoa, Choanoflagellida and Opalozoa.

Together, these compositional patterns support an interpretation of the recurrent community types as distinct ecosystem states characterized by contrasting trophic organization and productivity. WSC surface succession progressed from primary production and grazing during bloom development toward heterotrophy, parasitism and recycling later in the season, whereas deeper communities were consistently associated with organic matter processing and parasitism. This interpretation is consistent with previous Fram Strait observations linking surface communities to the processing of fresh phytoplankton-derived organic matter and deeper photic-zone communities to more complex and reworked substrates[7,18]. Importantly, these functional interpretations are based on relative sequence abundances and should not be interpreted as changes in absolute biomass; this distinction is particularly important for the low-chlorophyll, under-ice Type 1 community.

Core- and Indicator Taxa of the community types

We next assessed the compositional coherence of the eight community types by comparing within-type Jaccard distance distributions with the distribution across all samples (Supplementary Fig. SFgClustPurity). All types showed narrower within-type distributions than the global distribution, but their degree of homogeneity varied substantially. Type 2, associated with bloom onset in the MIZ, was among the most heterogeneous, with only a 7% reduction in mean pairwise distance and a 75% overlap with the global distribution. Along the WSC surface succession, compositional homogeneity increased progressively from Type 2 to Type 5, while a similar pattern emerged in deeper communities from Type 7 to Type 8. Type 5 and Type 8 were the most compositionally coherent states, with mean Jaccard distances reduced by 32% and 30%, respectively, and overlaps of only 26% and 23% overlap with the global distribution.

This pattern revealed a striking decoupling between diversity and compositional coherence. The least homogeneous communities, particularly Types 2 and 6, were associated with the highest chlorophyll-*a* concentrations, whereas the most compositionally coherent states occurred in early autumn, represented by Type 5 in the upper WSC and Type 8 at depth. Rapid bloom development may permit different taxa to dominate under contrasting years or local conditions, increasing compositional variability[62]. Conversely, autumn communities became more compositionally coherent despite increasing taxonomic diversity towards autumn and with depth [19]. This apparent paradox may arise because more diverse communities share a larger common pool of taxa, reducing the influence of the loss or gain of individual taxa on community dissimilarities [63]. Thus, seasonal succession was accompanied not only by changes in community composition and diversity, but also by a progressive tightening of community states around increasingly coherent compositional configurations.

To identify core taxa characteristic of individual community types, we selected taxa with mean relative abundance >0.01 that occurred in >75% of samples assigned to at least

one type (Supplementary Fig. SCoreSpec). This species-level analysis refined the phylum-level patterns (Fig. 5) and resolved distinct taxon-level associations with recurrent community states and their environmental niches.

Cold- and ice-associated communities were characterized by *Micromonas polaris* and *Fragilariopsis cylindrus*, species well known for their occurrence in cold waters related to sea-ice[64,65]. They reached their highest relative abundances in the cold EGC Type 1 and MIZ-associated Type 2 communities. *Phaeocystis pouchetii*, by contrast, was prevalent across all types and peaked in the highly productive Types 2 and 6, declining in late-season WSC and deep communities (Types 4, 5 and 8), in line with previous observations of a dominance of this species in Fram Strait during summer[54]. Heterotrophic taxa likewise showed distinct community-type associations. Cercozoa, known for their heterotrophic feeding on bacteria and microalgae[66], were most abundant in Types 2–3, consistent with active grazing during bloom development, whereas Type 6 was enriched in Picozoa and the ciliate *Leegaardiella*, indicating a distinct heterotrophic assemblage in Atlantic-influenced waters. Syndiniales, including Dino Groups I and II, occurred across most community types but were less prominent in the highly productive Type 2. Radiolaria were predominantly associated with low-chlorophyll communities, including EGC Type 1, late-season WSC Type 5 and deep WSC Types 7–8.

Taxonomic composition as predictor for chlorophyll

Because environmental conditions predicted community type and community type was strongly associated with chlorophyll-a concentration, we asked whether absolute chlorophyll-a concentrations could be inferred directly from relative taxon abundances. We started with the core taxa selected in the previous step and used a linear model to predict chlorophyll concentration:

$$\mathrm{Chl}_s^* = \beta_0 + \sum_i \beta_i \, p_{i,s}^*$$

where Chl_s^* is chlorophyll-*a* concentration and $p_{i,s}^*$ is relative abundance of taxon i in sample s, both fourth-root transformed to reduce right skewness. To identify indicator taxa most strongly associated with chlorophyll-a, we applied a two-stage feature-selection procedure. Taxon abundances were renormalized at each stage to sum to one within the candidate set, making model predictions independent of the abundance of the remaining community. The selection procedure reduced the core set to 20 indicator taxa. We then fitted the final model to the training data and evaluated its performance on an independent test set.

(both fourth-root transformed to reduce right skewedness of Chl_s and $p_{i,s}$ values) for and. The abundances of taxa, $p_{i,s}$ were always renormalized to sum to one making predictions independent of the abundance of the remaining community. Applying a two-stage feature selection procedure we reduced this set to 20 indicator taxa (see Methods) and then we fitted the model on training data and evaluated its performance on an independent test set.

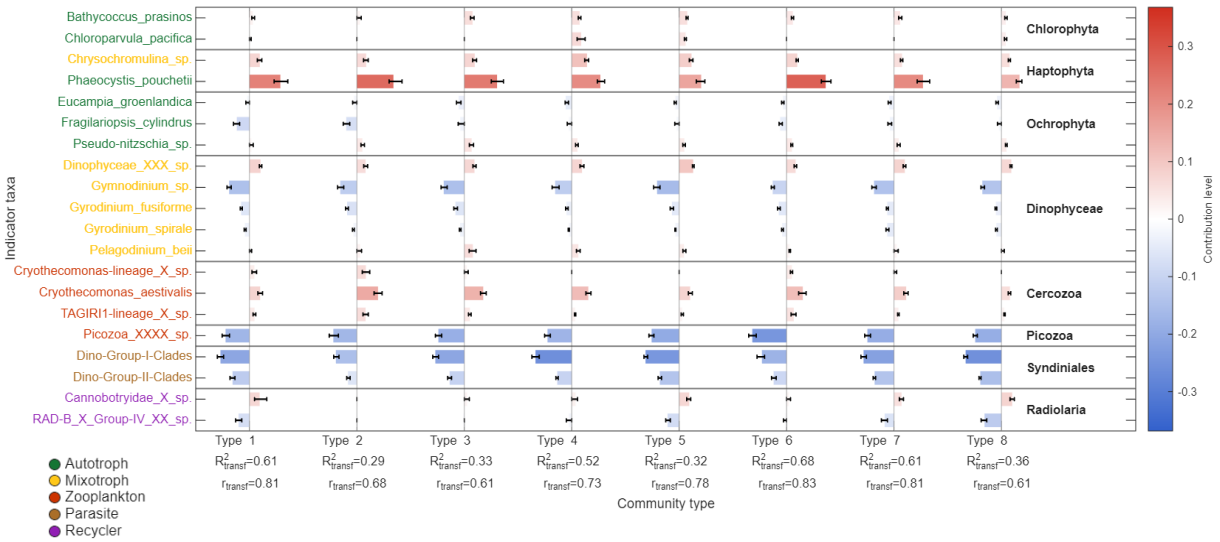


Fig. 6: Contributions of indicator taxa to chlorophyll-a prediction across community types.

Horizontal bars show the median contribution of each taxon to predicted chlorophyll-a concentration in the final linear model. Black whiskers indicate the interquartile range (25th–75th percentiles). Bar colour indicates the magnitude and direction of the contribution. Taxon names are coloured according to their functional role (see legend). Values below each community type indicate the coefficient of determination R^2_{transf} and Pearson correlation (r_{transf}) between observed and predicted chlorophyll-a concentrations, calculated for the fourth-root transformed response variable, $\text{Chl}^*_s = \text{Chl}^{0.25}_s$.

Predictive performance varied among community types (Fig. 6). Test-set R^2_{tranf} ranged from 0.29 to 0.68 and was highest for Types 1, 4, 6 and 7 ($R^2_{transf} \geq 0.52$)), whereas Types 2 and 5 showed the lowest explained variance ($R^2_{transf} = 0.29$ and 0.32, respectively). Nevertheless, observed and predicted chlorophyll remained strongly correlated within all community types ($r_{transf} = 0.61 \ldots 0.83$). Thus, although the unified model did not capture all within-type variation, it consistently recovered the main chlorophyll gradients from the relative abundances of indicator taxa.

To interpret these predictions, we calculated the median contribution of taxon i within each community type j as

$$b_{ij} = \beta_i, \text{median}_{s \in C_j}\, p^*_{i,s}$$

where β_i is the taxon-specific coefficient and C_j denotes the set samples assigned to community type j. This decomposition revealed distinct positive and negative taxonomic associations with predicted chlorophyll-a concentration (Fig. 6).

Phaeocystis pouchetii showed the strongest and most consistent positive contribution to predicted chlorophyll-*a*, followed by *Chrysochromulina* sp. and, more weakly, the chlorophytes *Bathycoccus prasinos* and *Chloroparvula pacifica*. Several abundant primary producers, known to dominate Arctic phytoplankton communities during

summer[67], including *Chaetoceros gelidus* and *Micromonas polaris*, were not retained as predictors in the final model, while *Fragilariopsis cylindrus* was negatively associated with chlorophyll-*a*. However, rather than indicating direct negative effects on productivity, these associations likely reflect importance of these taxa as zooplankton prey and their prevalence in cold, ice-influenced western communities. The generally weak or negative contributions of Ochrophyta taxa likewise contrasted with the strong positive signal of Haptophyta, particularly *Phaeocystis pouchetii*, a major primary producer in Fram Strait that is also relatively avoided by grazers[68]. This illustrates how phytoplankton concentrations are shaped by complex interactions between primary producers, environmental conditions and higher trophic levels, supporting the need for community-based approaches to their prediction.

Dinophyceae showed contrasting species-level associations, with an unidentified taxon contributing positively, whereas *Gymnodinium* sp. and several *Gyrodinium* taxa contributed negatively. Among heterotrophs, Cercozoa, particularly *Cryothecomonas aestivalis*, were positively associated with chlorophyll-*a*, consistent with their occurrence in bloom-associated communities, whereas Picozoa showed one of the strongest negative associations. Parasitic Syndiniales (grouped at the order level as Dino Groups I and II because of their high diversity) also showed consistently negative associations. Radiolaria showed contrasting patterns: RAD-B_X_Group-IV was negatively associated with chlorophyll-*a*, whereas Cannobotryidae showed a weak positive contribution.

Together, these results quantitatively demonstrate that taxonomic composition contains predictive information about ecosystem productivity, with haptophytes and bloom-associated consumers associated with high-chlorophyll states, whereas heterotrophic, parasitic and recycling-associated taxa characterise lower-chlorophyll communities. Importantly, these contributions represent indicator associations rather than direct effects on chlorophyll-*a*. In particular, although the genus Chaetoceros contributes substantially to phytoplankton biomass during blooms [36,54], this ecological importance was not reflected in the indicator set identified by our model. The negative association of *F. cylindrus* may reflect its affinity for ice-influenced, low-chlorophyll waters, whereas contrasting signals among diatoms and other producers could additionally reflect differences in grazing pressure or prey selectivity. The positive association of Cercozoa, in turn, may reflect their rapid trophic response and high competitive ability under low food concentrations [42,69]. Such mechanisms, however, cannot be resolved from the present predictive model alone. Our results extend the use of eDNA from describing community–environment associations to quantitatively predicting an ecosystem-level property from taxonomic composition.

Seasonally cumulated community productivity

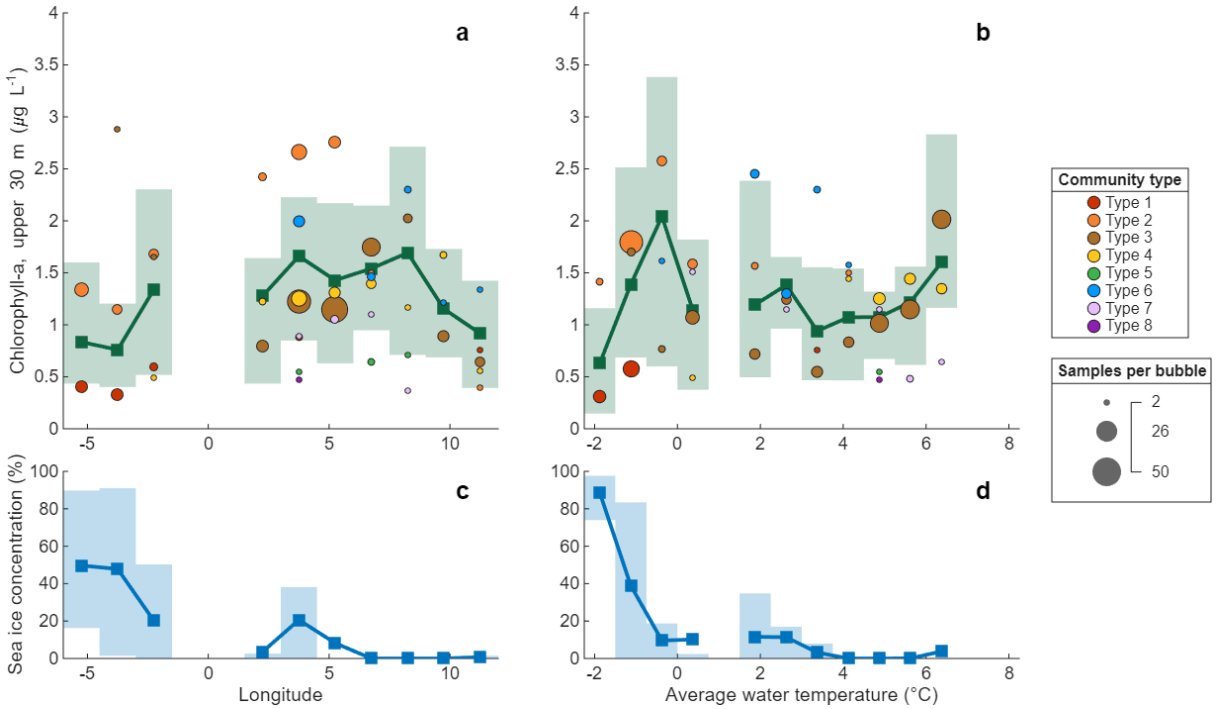


Fig. 7: Summer-averaged distributions of chlorophyll-a and sea ice along longitude and temperature gradients in Fram Strait.

a, b, Mean profile-averaged chlorophyll-*a* concentration in the upper 0–30 m from June to August, grouped by longitude **(a)** and mean water temperature **(b)**. Green lines and squares show seasonal means, and shaded areas indicate the 20th and 80th percentiles of profile-averaged chlorophyll-*a* concentrations. Coloured circles show mean chlorophyll-*a* concentrations for community types within each bin; circle size is the number of samples. **c, d,** Mean sea-ice concentration grouped by longitude **(c)** and mean water temperature **(d)**; blue lines and squares show seasonal means, and shaded areas indicate the 20th and 80th percentiles.

To assess how community succession translated into overall regional productivity patterns, we binned chlorophyll-a concentrations by longitude and temperature and averaged them from June to August. Productivity in this period was lowest in the western EGC, west of 2° W, and highest in the Atlantic-influenced waters east of 2° E (Fig. 7a). The observed productivity gradient is consistent with previous observations of Fram Strait, which show generally higher chlorophyll-*a* and primary production in Atlantic-influenced WSC waters than in the cold, ice-covered EGC [42,54,70]. Along the temperature gradient, chlorophyll-*a* exhibited a pronounced bimodal distribution, with distinct maxima near 0 °C and 6.5 °C and lower concentrations below −1 °C and between these optima (Fig. 7b). The first maximum was dominated by Type 2 communities and is consistent with the highly productive marginal-ice-zone regime, where sea-ice concentration drops below 20% (Fig. 7d), and water stratification promotes phytoplankton growth. The second maximum corresponds to warm Atlantic-influenced waters, where solar warming and Atlantic Water properties support Types 3 and 6. The bimodal response therefore reflects

the coexistence of two distinct hydrographic regimes capable of sustaining high phytoplankton biomass across Fram Strait but associated with different community states.

Our findings suggest that continued warming of Polar Water in the EGC could progressively reorganize the productivity landscape of Fram Strait. The lowest chlorophyll-*a* concentrations occurred in the coldest EGC waters (Type 1), whereas higher productivity was associated with two distinct thermal regimes near 0 °C (Type 2) and 6.5 °C (Type 3, 6). Continued environmental change is supported by observed warming of the EGC [71] and projected Arctic warming and Atlantification [72]. As Polar Water warms and sea ice retreats, conditions currently characteristic of the low-productivity western EGC may increasingly shift away from the cold, ice-covered state toward warmer conditions associated with greater phytoplankton biomass, potentially redistributing productive habitats and recurrent community types. However, warming alone is unlikely to determine future productivity: the two productivity maxima arise under contrasting hydrographic regimes, and their persistence will depend on how warming interacts with sea-ice loss, stratification, light and nutrient supply. Thus, a warmer Fram Strait may not simply become more productive, but may shift the balance among distinct high-productivity regimes, altering patterns of primary production, trophic transfer and carbon cycling.

Conclusions

Our unique 15-year eDNA Arctic time-series reveals that the high taxonomic complexity of eukaryotic microbial communities in Fram Strait is organized within a low-dimensional ecological manifold comprising recurrent whole-community states. Diffusion mapping recovered coherent seasonal, vertical and hydrographic structure, while community distribution modelling showed that these states can be predicted from a small set of variables representing seasonality, depth and hydrography. Community succession therefore emerges not as independent turnover of individual taxa, but as coordinated transitions among recurrent ecological states.

These states also link community composition to ecosystem function. Distinct community configurations were associated with contrasting productivity regimes, while similarly high chlorophyll-*a* occurred in compositionally distinct marginal-ice-zone and Atlantic-influenced states. Taxonomic eDNA composition itself contained sufficient information to quantitatively predict chlorophyll-a, demonstrating that eDNA can provide information on ecosystem function beyond taxonomic description alone. The strong association of community structure with temperature further suggests that continued warming and Atlantification could redistribute ecological states and reorganize the spatial and seasonal pattern of productivity across Fram Strait.

Our diffusion-map-based community modelling approach shifts the focus from fixed geographic units such as the EGC and WSC to biologically defined community states whose boundaries shift with environmental conditions. With a small set of predictors,

including temperature, seasonal progression, depth and longitude, our framework allows us to estimate the probabilities of occurrence of community types and anticipate transitions among them. This provides a basis for tracking how ecological states and associated productive zones shift through time and space, and for quantifying their redistribution under climate variability and change. In brief, our work moves from describing recurrent taxonomic patterns to a low-dimensional, environmentally predictable representation of whole-community states that links microbial composition to ecosystem productivity and provides a framework that makes use of long-term eDNA observations to track the redistribution of ecological states under climate change.

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Type	*Where and When*	*Chl a*	*Dominant Tax Groups*	*Distinctive Features and env. niche*
1. Arctic low-productivity	EGC, ice-covered region; surface and deep layers; June-September (absent in surface layer in July)	Very low	Small phytoplankton (Micromonas, Cryptophyta); Telonemia; Ciliophora; Sagenista; Radiolaria; Pseudofungi	Most 'Arctic' community; high share of heterotrophs, parasites, and recyclers.
2. Early-summer central bloom	Central grid; upper layers; June	Very high	Micromonas; Phaeocystis; Chaetoceros; Cercozoa; Ciliophora; heterotrophic Dinophyceae	Peak chlorophyll; low diversity except Cercozoa; minimal parasite fraction
3. Mid-summer eastern bloom	Eastern grid; surface; July	High	Chlorophyta; Haptophyta; Ochrophyta; Cercozoa; Ciliophora; mixotrophic Dinophyceae	Transitional stage between early bloom and late-summer communities; rise of mixotrophs
4. Late-summer mixed	Entire region; upper layers; mid-August–mid-September	Moderate	Primary producers remain; Telonemia; Opalozoa; decline of Cercozoa and Ciliophora; increase of Syndiniales	Onset of parasitism and heterotrophic control
5. Autumn pico-phytoplankton	Mainly east/center; upper layers; late September	Moderate-Low	Bathycoccus; Chloroparvula; Cryptophyta; Picozoa; Oomycetes; Syndiniales; Radiolaria	Shift to pico-phytoplankton; high parasite and recycler fractions
6. Early-summer WSC (upper + deep)	Early summer; WSC; upper and deep layers	Moderate–high	Haptophyta (Phaeocystis, Chrysochromulina); Chaetoceros; pico-zooplankton	Atlantic-influenced community with strong nanophytoplankton component
7. Deep heterotrophic (early–mid summer)	Deep layers; mid-June–August	Low	Picozoa; Choanoflagellida; Ciliophora; heterotrophic Dinophyceae; parasites; recyclers	Deep heterotrophic system with elevated Dinophyceae
8. Deep autumn recycler community	Deep layers; August–late September	Low	Picozoa; Ciliophora; Opalozoa; Syndiniales; Radiolaria (maximum)	Highest relative Radiolaria abundance (~10× mean); strong parasitic component

Table 1: Characteristic types of summer – early autumn communities in Fram strait.

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Author contribution statements

A.R. designed and performed the modelling and data analysis, processed the data and wrote the manuscript. K.M. designed the analysis, prepared the data and wrote the manuscript. R.V.-K., B.B. and T.G. contributed to the conceptual development of the study. R.McP. prepared the hydrographic data, and S.T. prepared the nutrient data. All authors contributed to structuring the manuscript and improving its quality, and approved the final version.

Competing interests

The authors declare no competing interests. The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript.

Methods

Data

Processed data comprised 882 samples collected during 13 expeditions of RV Polarstern and 4 expeditions of RV Maria S. Merian to the LTER HAUSGARTEN observatory in Fram Strait between 2009 and 2023. The number and location of sampled stations varied among years depending on sea-ice conditions and ship-time availability (Supplementary Table SCruises). Harmonized data and analysis scripts are available online at https://github.com/AlexRyabov/CommTypes/. Environmental data were compiled from multiple sources, processed and harmonized to associate each biological sample with the closest available physical and biogeochemical observations in space and time. The data sources are summarized below.

In situ samples during cruises

Ship-based hydrographic profiles of conductivity-temperature-depth (CTD) were recorded with a dual duct Sea-Bird 911+ (Sea-Bird Scientific, Washington, DC, USA) and averaged into 1 m bins. Conductivity and oxygen sensors were calibrated using water samples analysed on board with an Optimare Precision salinometer and titration, respectively. Chlorophyll-a fluorescence was also measured by the CTD. Seawater samples were collected using a rosette sampler equipped with 24 12-L Niskin bottles mounted on the CTD. Sampling depths ranged from the surface to 100 m and typically included 5, 10, 20, 50 and 100 m, together with the depth of the chlorophyll-a fluorescence maximum identified during the downcast. Samples were collected during the upcast after a waiting of time of 120 seconds before the Niskin bottle was closed. Sampling was conducted annually from June to September, with most samples obtained in June and July.

For chlorophyll-a measurements, 0.5–2.0 L of seawater were filtered onto 25-mm GF/F filters (Whatman, UK). For eDNA analyses, 2-L subsamples were transferred from Niskin bottles into PVC bottles [73]. Samples were sequentially filtered through 10, 3 and 0.4 µm, 45-mm Isopore membrane filters using a Millipore Sterifil filtration system (Millipore, USA) at 200 mbar, and DNA was extracted from material retained on the 0.4-µm filters.

To minimize cross-contamination among stations and depths, separate bottles and filtration units were used for each sampling depth. After each station, sampling bottles and filtration equipment were thoroughly rinsed with freshwater and, before sampling at the next station, bottles were additionally rinsed with water from the corresponding Niskin bottle.

DNA isolation. Genomic DNA was extracted from field samples according to the manufacturers' protocols using either the NucleoSpin Plant Kit (Macherey-Nagel, Germany; until 2015) or the DNeasy PowerWater Kit (Qiagen, Germany; thereafter). Both kits have been shown to provide comparable characterization of marine 18S rRNA gene assemblages, suggesting that this methodological change was unlikely to substantially affect the long-term community patterns resolved in our dataset [74]. DNA was extracted separately from each size fraction and eluted in 60 µl of the kit-specific elution buffer.

Extracts from the different size fractions were subsequently pooled for sequencing. DNA concentrations in the pooled samples ranged from 0.2 to 20 ng μl^{-1}, and extracts were stored at −80 °C until further processing.

18S rDNA amplification and Illumina sequencing. A fragment of the 18S rRNA gene encompassing the hypervariable V4 region was amplified using the primers 528iF (GCGGTAATTCCAGCTCC) and 964iR (ACTTTCGTTCTTGATYRR) [46]. PCR reactions were performed in a final volume of 50 µl containing 0.02 U Phusion Polymerase (Thermo Fisher Scientific, Germany), 5 µl 10× polymerase buffer, 0.8 mM of each dNTP (Eppendorf, Germany), 0.2 µM of each primer and 1 µl template DNA. Following initial denaturation at 94 °C for 2 min, 35 amplification cycles were performed, each consisting of denaturation at 94 °C for 20 s, annealing at 58 °C for 30 s and extension at 68 °C for 30 s, followed by a final extension at 68 °C for 10 min. PCR products were purified from 1% [w/v] agarose gels using either the NucleoSpin Gel Kit (Macherey-Nagel, Germany) or the MinElute PCR Purification Kit (Qiagen, Germany). DNA concentrations were measured using a Quantus Fluorometer (Promega, USA), and purified amplicons were diluted in TE buffer to 0.2 ng μl^{-1} before library preparation. Libraries were prepared following the Illumina 16S Metagenomic Sequencing Library Preparation protocol and sequenced on an Illumina MiSeq platform. Raw reads were approximately 250 bp long. Sequence data generated in this study were deposited through GFBio [75] in the European Nucleotide Archive (ENA) under the following project accession numbers: 2009, PRJEB106806; 2010, PRJEB108217; 2011, PRJEB107315; 2012, PRJEB108221; 2013, PRJEB106980; 2014, PRJEB107317; 2015, PRJEB107320; 2016, PRJEB108222; 2017, PRJEB108240; 2018, PRJEB108638; 2019, PRJEB107225; 2020, PRJEB107226; and 2021, PRJEB108237; data from 2022 and 2023 will be uploaded upon acceptance.

Sequence processing and taxonomic annotation. Raw reads were processed into amplicon sequence variants (ASVs) using DADA2 v.1.18 in R v.4.0[76]. Forward and reverse primers were removed using Cutadapt v.3.4 [77], and low-quality 3' ends were trimmed based on visual inspection of quality profiles [78]. Reads were filtered based on their expected error rates to ensure a minimum sequence quality. Sequencing errors were modeled and corrected using the DADA2 error model [76]. Forward and reverse reads were merged with a minimum overlap of 25 bp and no mismatches. Chimeric sequences were subsequently identified and removed using DADA2 [76]. The resulting ASVs were taxonomically annotated against 18S rRNA gene sequences in the Protist Ribosomal Reference database (PR2, v.4.14; [79]). Under the PR2 nomenclature, unresolved assignments are indicated by suffixes corresponding to the highest resolved taxonomic rank: `_sp` for genus, `_X_sp` for family, `_XX_sp` for order, `_XXX_sp` for class and `_XXXX_sp` for phylum.

Metazoa were excluded to focus the analysis on protists and other unicellular eukaryotes. Rare taxa were removed using two simultaneous criteria: taxa had to occur in more than 5% of samples and have a median relative abundance greater than 1×10^{-4} among samples in which they were present, based on an initial within-sample normalization. After filtering, ASV abundances were renormalized within each sample to

obtain the final relative-abundance matrix. Only samples collected at depths ≤ 102 m were retained, resulting in the exclusion of eight deeper samples collected in 2020.

For broad comparisons of community composition among community types, ASV abundance and richness were aggregated at the phylum level. Dinoflagellates were treated separately because of their high diversity (>100 ASVs) and contrasting ecological roles. We therefore divided them into Dinophyceae sensu stricto and the predominantly parasitic Syndiniales, while all other taxa were analysed at the phylum level. This yielded 15 taxonomic groups representing major ecological roles (Fig. 4): primary producers (Chlorophyta, Haptophyta, Ochrophyta and Cryptophyta), mixotrophs (Dinophyceae), consumers (Picozoa, Choanoflagellida, Telonemia, Opalozoa, Cercozoa and Ciliophora), parasites (Pseudofungi and Syndiniales), and organic matter processors (Sagenista and Radiolaria). The final dataset comprised 882 samples and 1,301 ASVs assigned to these 15 groups.

For the identification of core and indicator taxa (Figs. fgChlTypes and SCoreSpec), most ASVs were aggregated at the species level. ASVs that could not be resolved to species level were retained at their highest resolved taxonomic rank (e.g. Picozoa_XXXX_sp). Because of the high taxonomic diversity of Syndiniales, these ASVs were instead aggregated at the order level into Dino-Groups I and II.

Environmental factors

Satellite data

Monthly surface photosynthetically available radiation (PAR), the diffuse attenuation coefficient at 490 nm (Kd490), and sea-surface temperature (SST) were obtained from MODIS Aqua Level-3 mapped products distributed by the NASA Ocean Biology Processing Group (OBPG). PAR (Einstein m^{-2} d^{-1}) and Kd490 (m^{-1}) were obtained from the R2022.0 products (DOIs: 10.5067/AQUA/MODIS/L3M/PAR/2022 and 10.5067/AQUA/MODIS/L3M/KD/2022), and SST (°C) from the MODIS Aqua Global Level-3 Mapped SST product v2019.0 (DOI: 10.5067/MODSA-MO4D9). All products were monthly means at approximately 4-km spatial resolution.

PAR at sampling depth z was estimated from surface PAR and Kd490 assuming exponential light attenuation according to the Beer–Lambert law:

$$PAR(z) = PAR_0 \exp(-K_{d490} \cdot z) \ ,$$

where PAR_0 is satellite-derived surface PAR and z is sampling depth in metres.

Ice

Monthly mean sea-ice concentration (SIC, %) was obtained from https://www.meereisportal.de/ and used to characterise average ice conditions at each sampling location. Daily SIC was obtained from the Sea Ice Data Center, https://data.seaice.uni-bremen.de/, based on ASI satellite products (Spreen et al., 2008; Melsheimer & Spreen, 2019, 2020). For each sample, we extracted three

complementary measures of sea-ice conditions at the corresponding location: monthly mean SIC to represent background ice conditions, daily SIC on the sampling date to represent concurrent conditions, and daily SIC 14 days before sampling to represent recent ice history. The latter was included to test whether preceding ice conditions contributed to variation in community composition.

CTD

CTD measurements, chlorophyll-a concentrations and nutrient data were obtained from the PANGAEA database; the complete list of datasets is provided in the Supplementary Data Sources. CTD profiles were averaged at 1-m vertical resolution, and each biological sample was assigned the temperature measured within ±5 m of its sampling depth. Missing temperature values (2.7% of the dataset) were imputed as described below.

Mixed layer depth

We estimated mixed layer depth using a density-threshold approach conceptually similar to [80] based on potential density calculated from CTD measurements. Instead of using the density at the shallowest measurement, we defined the reference density as the median potential water density σ_θ within the upper 10 m and identified the first depth below 10 m where σ_θ exceeded this reference by 0.1 kg m^{-3}. This procedure provides a robust estimate of the mixed layer depth while reducing sensitivity to near-surface noise.

Nutrients

Concentrations of nitrate, nitrite, ammonium, silicate and phosphate were measured during the same cruises as the biological sampling. DIN was calculated as the sum of nitrate, nitrite and ammonium. Because nutrient samples were collected at depths that did not always coincide with biological sampling depths, vertical nutrient profiles were interpolated at 1 m resolution (function interp1, method pchip, MATLAB), and the value at the depth closest to each biological sample was used. To test whether community composition was related to the nutrient reservoir below the euphotic zone, we calculated mean DIN between 100 and 200 m ($DIN_{100-200}$) from the interpolated vertical profiles. However, none of the nutrient-related variables was retained as a direct predictor of community composition during predictor selection. Nutrient measurements were missing for 42.4% of samples; these values were therefore imputed to obtain a complete dataset for subsequent analyses. Imputation was based on depth, geographic location and sampling date using boosted regression trees. Model hyperparameters were tuned by leave-one-year-out cross-validation, with each sampling year treated as a separate fold. Observed nutrient values were retained unchanged, and only missing entries were replaced by model predictions.

Data imputation

Environmental measurements were incomplete across cruises, particularly for nutrients. To maximise the number of samples available for downstream analyses, missing values were imputed using boosted regression trees with day of year, longitude,

latitude and sampling depth as predictors. The resulting estimates can therefore be interpreted as nonlinear spatiotemporal predictions based on the location, depth and date of each sample. Machine-learning models were evaluated using both the original observations and the imputed datasets. Overall, 2.7% of temperature values and 42.4% of nutrient values were imputed.

Decimal month

For visualisation of seasonal patterns, day of year (DOY) was converted to decimal month as $\mathrm{Month}=1+12(DOY-1)/365$ which, for a non-leap year, ranges from 1 on 1 January to approximately 12.97 on 31 December.

Deviation index

Because baseline abundance and richness differed substantially among taxonomic groups, we normalised each metric by its group-specific median and defined a deviation index as the logarithm of this ratio. Positive values indicate enrichment relative to the reference level, whereas negative values indicate depletion; values of +1 and −1 correspond to tenfold enrichment and depletion, respectively.

For each sample j and taxonomic group k we calculated total relative abundance as

$$S_{kj} = \sum_{i \in k} p_{ij} \,,$$

where p_{ij} is the relative abundance of taxon i in sample j. Group richness was calculated analogously as the number of taxa present,

$$S_{kj}^{R} = \sum_{i \in k} \mathbf{1}\,(p_{ij} > 0) \,,$$

Zero S_{kj} values were replaced with one-tenth of the smallest positive value in the complete grouped matrix. For each metric and taxonomic group k, we defined the reference level S_k^* as the median S_{kj} across all samples. We then calculated the logarithmic deviation of S_{kj} from the reference level as

$$D_{kj} = \log_{10}\left(\frac{S_{kj}}{S_k^*}\right)$$

and analogously for richness.

Because the index is based on relative abundances, it should not be interpreted as an absolute measure of biomass, absolute abundance, or growth rate. For example, under low-light conditions, growth and absolute abundance may decline across many taxa, but autotrophs are likely to be more strongly affected than heterotrophs. This can lead to

a relative depletion of autotrophs ($D_a < 0$) and a relative enrichment of heterotrophs ($D_h > 0$). Thus, even if heterotrophs do not increase in absolute abundance, a positive D_h value indicates their increased relative contribution to the community in sample j.

Statistical analysis

Diffusion maps

We used diffusion-map embedding to obtain a low-dimensional representation of variation in community composition (Diffusion Maps package, MATLAB). Diffusion maps are a nonlinear manifold-learning method that projects objects defined by pairwise similarities into a low-dimensional space while preserving the local geometry of the similarity network [27,28]. Detailed description of our procedure is found in [29]. Unlike conventional ordination, diffusion maps are built from a network of trusted links between nearest neighbours while links between dissimilar communities are excluded. By preserving the local geometry of the network connecting similar communities, the method defines distances between remote communities as the diffusion time across the network of trusted links. Pairwise community similarity was quantified using the Jaccard index based on ASV occurrence, such that communities with similar taxonomic composition are located close together in diffusion-map space.

The number of trusted links between communities was selected using environmentally informed parameter tuning, choosing the value that maximised the association between the resulting diffusion-map coordinates and environmental variables; this procedure yielded 60 neighbours. To assess whether the inferred gradients were driven by temporal or spatial autocorrelation among samples collected close in space and time, we performed a sensitivity analysis in which the diffusion map was constructed using only pairwise similarities between samples collected in different years.

Correlation dimension

We estimated the correlation dimension from the 10-dimensional diffusion-map representation to quantify the effective dimensionality of the community manifold. The correlation dimension is derived from the scaling of the correlation sum, $C(r)$, which measures the fraction of point pairs separated by less than a distance r, such that $C(r) \propto r^D$ over the scaling region [31]. The exponent D therefore provides an estimate of the number of effective dimensions required to describe the local structure of the community distribution.

Community types

We identified recurrent community states by clustering samples in diffusion-map space using the first six diffusion-map coordinates. Gaussian mixture models were fitted to this representation using fitgmdist (MATLAB), allowing samples with similar ASV composition to form probabilistic clusters. Models with different numbers of mixture components were compared using the overall silhouette score, and the number of components maximising this criterion was selected, yielding 8 community types. Each

sample was assigned to the mixture component with the highest posterior probability. To assess within-type compositional coherence, we compared the distribution of pairwise Jaccard distances within each community type with the distribution across the complete dataset.

Machine-learning classification

To relate community types to their spatial, seasonal and environmental distributions, we trained two multiclass support vector machine (SVM) classifiers. The first model described where and when each community type occurred and considered latitude, longitude, DOY and sampling depth as candidate predictors. Sequential feature selection retained longitude, DOY and depth. The second model was designed to identify environmental variables associated with community composition and considered DOY, depth, temperature, salinity, nutrient concentrations (phosphate, silicate, nitrate and nitrite), and remotely sensed variables, including surface PAR, estimated PAR at sampling depth, SST, sea ice concentration, surface chlorophyll and the diffuse attenuation coefficient. The final environmental model retained temperature, DOY and depth. Models including temperature used 24 imputed values, corresponding to 2.7% of the dataset.

Community types were classified using multiclass SVMs implemented with fitcecoc (MATLAB), using one-versus-one error-correcting output codes and radial basis function kernels. Predictors were standardised before model fitting, and observations were weighted inversely to the number of samples in each community type to reduce the effect of class imbalance. Five-fold cross-validation was blocked by cruise and stratified by month, such that samples from the same cruise were assigned to the same fold while maintaining a broadly representative seasonal distribution across folds.

Predictors were selected using sequential feature selection (sequentialfs, MATLAB), with cross-validated misclassification rate as the selection criterion. SVM hyperparameters were first optimised for the full candidate predictor set, after which feature selection was performed and the BoxConstraint, C, and KernelScale, σ, parameters were re-optimised for the selected predictor subset. The combination yielding the lowest cross-validation error was $C = 18.33$ and $\sigma = 8.57$ for the longitude–DOY–depth model, and $C = 20.69$ and $\sigma = 10.25$ for the temperature–DOY–depth model. The same cross-validation folds were used for feature selection, hyperparameter tuning and calculation of the reported classification error and confusion matrix. Final ECOC models were fitted to the complete dataset, and their classification scores were calibrated to posterior class probabilities using MATLAB's built-in posterior-fitting procedure (FitPosterior, true). These probabilities were used to characterise community-type occurrence across environmental gradients.

Partial dependence was calculated using partialDependence (MATLAB). Two-dimensional partial-dependence surfaces were used to visualise predicted community-type distributions across environmental gradients and were additionally averaged across depth intervals for Fig. 3a–f. Model stability was assessed using one-dimensional

partial-dependence curves obtained from 100 bootstrap resamples of the original dataset; 95% bootstrap intervals are shown as shaded areas in Supplementary Fig. SfgNicheNoDOY.

Selection of taxonomic indicators for predicting chlorophyll-a concentrations

For each community type, we defined characteristic core taxa as those present in at least 75% of samples and with a mean relative abundance >0.01. This criterion identified 43 core taxa (Supplementary Fig. SCoreSpec).

These core taxa were used as the initial set of candidate indicator taxa. Samples were stratified by community type and split into training (70%) and independent test (30%) sets. All indicator taxon selection and model fitting were performed using the training set, while the test set was reserved exclusively for final model evaluation.

Indicator taxa were selected in two stages. At each stage, the relative abundance of taxon i in sample s was renormalised for the current set of candidate taxa and power-transformed to reduce right skewness in the abundance distributions:

$$p^*_{i,s} = \left(\frac{p_{i,s}}{\sum_j p_{j,s}}\right)^{0.25}, \text{where } j \in \text{candidate taxa}$$

Chlorophyll-a concentration was likewise transformed to reduce right skewness: $\mathrm{Chl}^*_s = \mathrm{Chl}_s^{0.25}$.

Models predicted transformed chlorophyll as

$$\mathrm{Chl}^*_s = \beta_0 + \sum_i \beta_i \, p^*_{i,s}$$

using sample weights $w_t = 1/N_t$, where N_t is the number of training samples assigned to community type t, thereby giving equal weight to each community type.

In the first stage, we fitted a ridge regression including all candidate core taxa, because it can accommodate correlated predictors by shrinking their coefficients rather than requiring their exclusion. Ridge models were fitted using the MATLAB function fitrlinear, with the regularization parameter λ selected by sixfold cross-validation. The contribution of each taxon to the prediction for each sample was defined as $\beta_i p^*_{i,s}$ and taxa were ranked by their mean absolute contribution across the training set. The 25 highest-ranked taxa were retained. In the second stage, this set was further reduced using stepwise linear regression in MATLAB, with AIC as the selection criterion and entry and removal thresholds of $p_{enter} = 0.05$ and $p_{remove} = 0.1$. The abundances of the retained taxa were then renormalized and transformed using the same procedure and used to fit the final weighted linear regression model (fitlm function in MATLAB). The resulting coefficients β_i are reported in Table SChlPred. The training and test performance of the final model is shown in Supplementary Fig. SChlTrainError and SChlTestError.

For visualization in Fig. 6, we calculated the mean and interquartile range (Q25–Q75) of each taxon's contribution separately for each community type. Positive and negative values indicate contributions towards higher and lower predicted chlorophyll-a concentrations, respectively.

Calculation of summer chlorophyll-a distributions along temperature and longitude gradients

To estimate mean summer chlorophyll concentration in the upper 0–30 m (Fig. 7), we used profiles collected from June to August that contained measurements at least two depths spanning the 10–20 m interval. Profiles were interpolated using (interp1, method pchip, MATLAB) onto a 1 m grid from 0 to 30 m; outside the observed depth range, the nearest measured value was retained. Mean chlorophyll concentration over 0–30 m and mean water temperature were then calculated for each profile. Profiles were grouped into longitude or temperature bins. Bins containing fewer than four profiles over the entire observation period were excluded. To reduce bias from uneven seasonal sampling, profile-averaged chlorophyll values were first averaged within 5-day intervals and the resulting means were averaged across the season. For each bin, the 20th and 80th percentiles of profile-averaged chlorophyll concentrations were also calculated. Sea-ice concentration was processed analogously.

To represent community types in the same plots, samples from the upper 0–30 m were grouped by bin and community type, and mean chlorophyll concentration and sample number were calculated for each group. Unlike the seasonal means, these values were calculated from individual samples rather than profiles and assigned to bins according to the longitude or mean water temperature of the corresponding profile. Thus, although seasonal means and community-type values are related, they are not directly comparable because a single vertical profile may contain samples assigned to different community types.